



RebootLoop

SECURELY UNLOCK THE NEXT LIFE OF STUDENT TECHNOLOGY

A privacy-first decision platform for repair, reuse, donation and responsible recycling

SHAH NOOR MOSTAFA BHUIYAN | NAFE IBNE MAMUN | ALEXANDER KAI CRYAN

THREE-STUDENT TEAM | SCHOOL OF COMPUTING | PITCH FOR THE PLANET 2026

40%

of Australians surveyed considered data security a barrier to recycling a mobile phone.

MobileMuster / IPSOS 2024 survey, n=2,000

WHY THIS MATTERS

The problem

511,000 t of e-waste generated in Australia in 2019

657,000 t projected annually by 2030 - almost 30% growth

A peer-reviewed study of Sydney university students found awareness of e-waste, but a severe lack of knowledge about collection points and current recycling programs.

Sources: Australian Government DCCEEW (2026); Islam, Dias & Huda (2021)

The missing layer

Existing services answer where to take a device. RebootLoop answers the harder questions first:

IS IT SAFE? CAN IT BE REUSED? WHAT NEXT?

UN SDG alignment

12

PRIMARY

Target 12.5
Reduce waste through prevention, reduction, recycling and reuse.

12.8

SUPPORTING

Provide the information and awareness needed for sustainable choices.

13

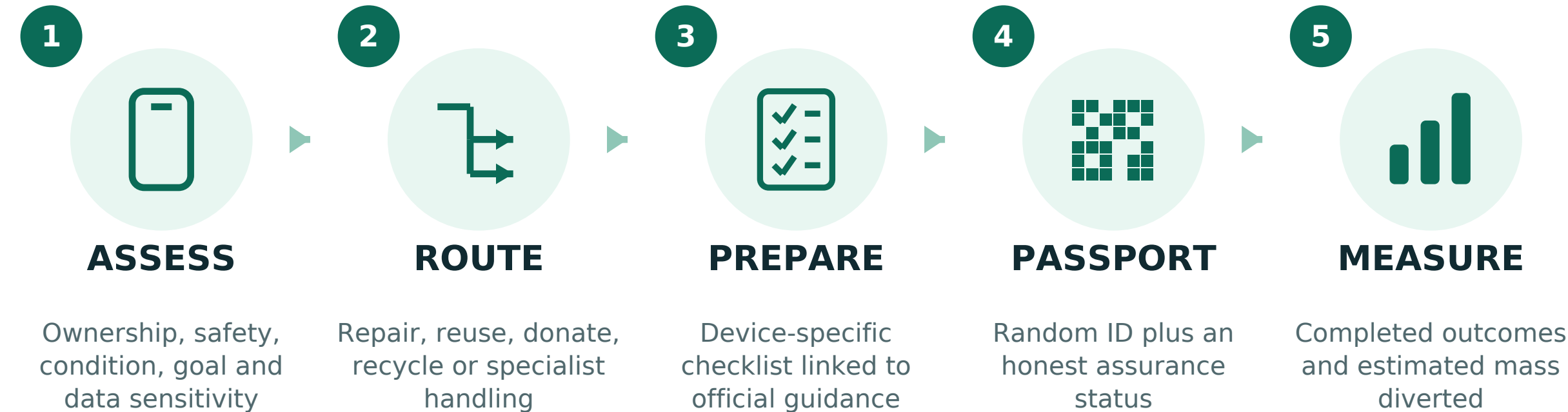
SUPPORTING

Extend device life and reduce demand for resource-intensive replacements.

HOW IT WORKS

One safe decision

UNDER 2 MINUTES | NO ACCOUNT | NO FILE UPLOAD | NO SERIAL NUMBER



SAFETY-FIRST ROUTING

Physical safety > authority and ownership > data sensitivity > device viability > circular outcome

TWO DEMO JOURNEYS

The logic handles the difficult cases

DEMO A

Working phone + donation

Owned | works | safe | everyday data

PREPARE, THEN DONATE
Official checklist -> self-declared passport

DEMO B

Laptop will not boot + highly sensitive data

Owned | non-functional | high sensitivity

PROFESSIONAL DESTRUCTION, THEN RECYCLING
No false claim that a reset occurred

INNOVATION + TECHNICAL EXCELLENCE

Why RebootLoop is different

- Transparent rules, not a mysterious AI score. Every recommendation shows exactly why it was selected.
- Privacy by design. The passport excludes names, student IDs, serial numbers, IMEI, account data and free text.
- Honest assurance. It separates guidance generated, self-declared preparation and future partner verification.
- Works with existing services. RebootLoop routes users to repair, donation and accredited recycling pathways instead of duplicating them.

MEASUREMENT

Impact we can defend

20-50 devices in an initial MQ discovery pilot

- Completed devices by route: repair, reuse, donation or recycling
- Estimated mass diverted using published category reference weights
- Confidence change before and after guidance
- Guided vs self-declared vs partner-verified outcomes

No invented carbon figure. Outcomes are labelled illustrative until piloted.

FEASIBLE + SCALABLE

A practical MQ pilot

Start with a one-day campus pop-up. RebootLoop records only category, route and assurance state - never personal files or hardware identifiers.

- MQ Sustainability and Facilities - host, communications, reporting
- MQ IT / security / privacy - guidance review and managed-device pathway
- Existing repair, donation and recycling services - verified endpoints

SCALE BY CHANGING LOCAL PARTNERS - NOT THE CORE SAFETY LOGIC

WHAT THE MVP DOES NOT CLAIM

Trust requires boundaries

- It does not erase data or guarantee non-recoverability.
- It does not certify destruction or endorse an unverified partner.
- Damaged batteries and managed devices stop the normal flow.

THE NEXT STEP

Pilot the decision layer at MQ - then measure whether confidence becomes completed circular action.
ONE SAFE DECISION. ONE MORE DEVICE IN CIRCULATION.